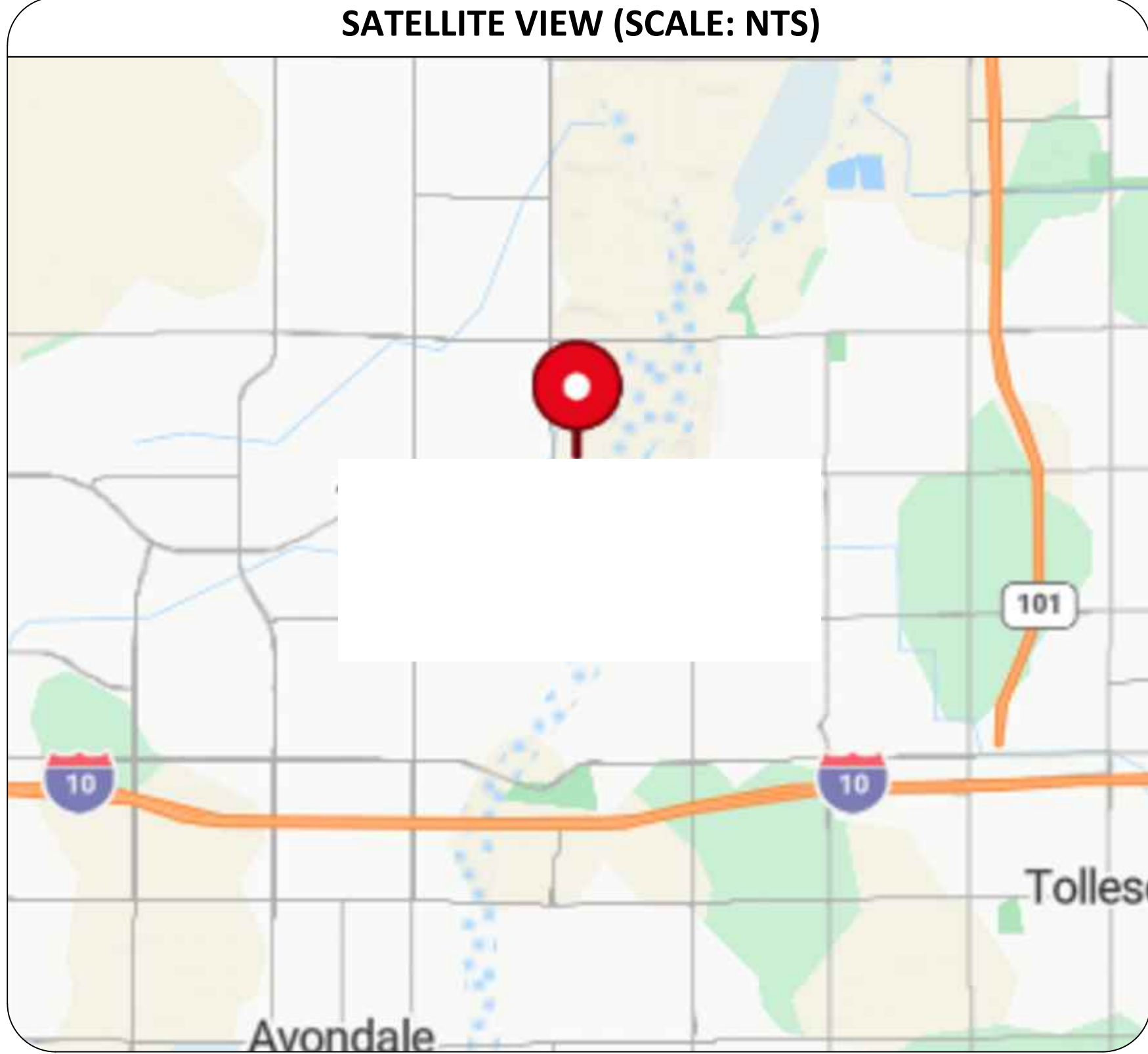


PROJECT: PV SYSTEM DETAILS	
[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(144) MODULES x 600W	(2) INVERTERS x 28000W
86.400 kW	56.000 kW



SCOPE OF WORK
- PROPOSED PROJECT TO INSTALL GRID- TIED PV ONLY SYSTEM ON AN EXISTING ROOFTOP STRUCTURE AT A COMMERCIAL LOCATION: 12345 STREET NAME, LITCHFIELD PARK, AZ 9874. COORDINATES: 33.495446, -112.320826 - THE PV MODULES WILL BE SECURED ON ROOF USING PRE-ENGINEERED ROOF RACKING SYSTEM. - INSTALLED PV SYSTEM WILL BE CONNECTED TO APARTMENT'S SERVICE EQUIPMENT USING APPLICABLE INTERCONNECTION METHOD (AS PER AHJ). - THE ENERGY PRODUCED BY PV SYSTEM WILL BE USED TO POWER THE ENERGY REQUIREMENTS. EXCESS POWER IS SENT BACK TO THE UTILITY GRID.

APPLICABLE GOVERNING CODES & NOTES
THIS PROJECT SHALL COMPLY WITH THE FOLLOWING CODES: 2021 INTERNATIONAL BUILDING CODE 2021 INTERNATIONAL FIRE CODE 2021 INTERNATIONAL RESIDENTIAL CODE 2020 NATIONAL ELECTRICAL CODE
ALL OTHER RELATED STATE & AHJ BY-LAWS & ORDINANCES

DESIGN CRITERIA
- CONSTRUCTION: COMMERCIAL - EXPOSURE CATEGORY : B, RISK CATEGORY : II - WIND SPEED (ASCE 7-16) : 101 MPH - GROUND SNOW LOAD (ASCE 7-16) : 0 PSF - ASHRAE EXTREME LOW: -1°C - ASHRAE HIGH TEMP (2% AVG.) : 43°C - ASHRAE DISTANCE ABOVE ROOF (7/8") : 65°C

SHEET INDEX	
PV-0	PROJECT SUMMARY
PV-1	SITE PLAN
PV-2	ROOF ATTACHMENT DETAILS
PV-3	BOM & STRING LAYOUT
S-1	STRUCTURAL DETAILED DIAGRAM
E-1, E-2	ELECTRICAL DIAGRAM & CALCULATIONS
E-3	PLACARDS
E.S	EQUIPMENT SPECIFICATIONS

EQUIPMENT DETAILS	
(#) PV MODULES	(144) JINKO SOLAR JKM600N-78HL4-BDV
(#) INVERTER	(2) GROWATT HYBRID WIT 28K-HU-US L2 (208V)
(#) OPTIMIZERS	(144) TIGO (TS4-A-O) OPTIMIZERS
OUTPUT CURRENT	171.20A
ROOF RACKING SYSTEM	S-5 PROTEA BRACKET /IRONRIDGE XR-10
STRING	12 STRINGS OF 12 MODULES
POINT OF INTERCONNECTION	LINE SIDE TAP IN THE EXTERIOR MAIN SERVICE PANEL WITH 400A MAIN BREAKER, 400A BUSBAR, 120/208V, 4W, 3ϕ
UTILITY	APS
AHJ	AVONDALE CITY

GENERAL NOTES
- ALL COMPONENTS ARE UL LISTED & NEC CERTIFIED, WHERE WARRANTED - A DISCONNECTING MEANS SHALL BE INSTALLED AT A READILY ACCESSIBLE LOCATION EITHER OUTSIDE OF THE BUILDING OR STRUCTURE OR INSIDE NEAREST POINT OF ENTRANCE OF THE SYSTEM CONDUCTORS. - HEIGHT OF THE INTEGRATED AC/DC DISCONNECT SHALL NOT EXCEED 6'-7" PER NEC CODE 240.24. - A GROUNDING ELECTRODE SYSTEM IN ACCORDANCE WITH NEC 690.47 AND 250.50 THROUGH 60 AND 250-166 SHALL BE PROVIDED. PER NEC GROUNDING ELECTRODE SYSTEM OF EXISTING BUILDING MAY BE USED AND BONDED TO THE SERVICE ENTRANCE. IF EXISTING SYSTEM IS INACCESSIBLE OR INADEQUATE A SUPPLEMENTAL GROUNDING ELECTRODE WILL BE USED AT THE INVERTER LOCATION CONSISTING OF A UL LISTED 8 FT. GROUND ROD WITH ACORN CLAMP. GROUNDING ELECTRODE CONDUCTORS SHALL BE NO LESS THAN #8 AWG AND NO LARGER THAN #6 AWG COPPER AND BONDED TO THE EXISTING GROUNDING ELECTRODE TO PROVIDE FOR A COMPLETE SYSTEM. - PHOTOVOLTAIC MODULES ARE TO BE CONSIDERED NON-COMBUSTIBLE. - PHOTOVOLTAIC INSTALLATION WILL NOT OBSTRUCT ANY PLUMBING. MECHANICAL, OR BUILDING ROOF VENTS. - WATER PROOF CONNECTORS AND HUBS SHALL BE USED WHERE APPLICABLE PER NEC 312.2 AND 314.15. - GROUNDING/BONDING BUSHINGS SHALL BE INSTALLED WHERE APPLICABLE PER NEC 250.92. - ALL EXTERIOR RACEWAYS ON WALLS SHOULD BE EMT UNLESS NOTED OTHERWISE. - CONTRACTOR SHALL INSTALL EQUIPMENT IN ACCORDANCE WITH MANUFACTURER INSTRUCTIONS & NOTHING IN THIS DOCUMENT SHALL BE INTERPRETED IN A WAY THAT OVERRIDES THEM. - CONTRACTOR IS RESPONSIBLE FOR VERIFICATION OF ALL CONDITIONS, DIMENSIONS & DETAILS IN THIS DOCUMENT. ALL DIMENSIONS SHALL BE VERIFIED BY SUBCONTRACTOR BEFORE INSTALLATION.
PHOTOVOLTAIC SYSTEM FIRE CLASSIFICATION LISTING IN ACCORDANCE WITH UL1703 STANDARD.

ELECTRICAL NOTES
- THE EQUIPMENT AND ALL ASSOCIATED WIRING AND INTERCONNECTION SHALL BE INSTALLED ONLY BY QUALIFIED PEOPLE. A QUALIFIED PERSON IS ONE WHO HAS SKILLS AND KNOWLEDGE RELATED TO THE CONSTRUCTION AND OPERATION OF THE ELECTRICAL EQUIPMENT AND INSTALLATIONS AND HAS RECEIVED SAFETY TRAINING TO RECOGNIZE AND AVOID THE HAZARDS INVOLVED. (NEC690.4(E) AND 705.6) - LOCAL UTILITY PROVIDER SHALL BE NOTIFIED PRIOR TO USE AND ACTIVATION OF ANY SOLAR PHOTOVOLTAIC INSTALLATION. FOR A LINE SIDE TAP CONNECTION, UTILITY NEEDS TO BE NOTIFIED WELL IN ADVANCE TO COORDINATE BUILDING ELECTRICAL SHUT OFF. - NEW CONDUIT ROUTING SHOWN IS ESSENTIALLY SCHEMATIC. SUBCONTRACTOR SHALL LAY OUT RUNS TO SUIT FIELD CONDITIONS AND THE COORDINATION REQUIREMENTS OF OTHER TRADES. - ARRAY WIRING SHOULD NOT BE READILY ACCESSIBLE EXCEPT TO QUALIFIED PERSONNEL. - ALL EXTERIOR CONDUIT, FITTINGS, AND BOXES SHALL BE WATERTIGHT AND APPROVED FOR USE IN WET LOCATIONS . - WIRING METHODS FOR PV SYSTEM CONDUCTORS AREN'T PERMITTED WITHIN 10 IN. OF THE ROOF DECKING OR SHEATHING EXCEPT WHERE LOCATED DIRECTLY BELOW THE ROOF SURFACE THAT'S COVERED BY PV MODULES AND ASSOCIATED EQUIPMENT WIRING - BACK-FED BREAKER MUST BE AT THE OPPOSITE END OF BUS BAR FROM THE MAIN BREAKER OR MAIN LUG SUPPLYING CURRENT FROM THE UTILITIES. - ALL CONDUCTORS AND WIRE TIES EXPOSED TO SUNLIGHT ARE LISTED AS UV RESISTANT. - CONTRACTOR SHALL FOLLOW ALL ELECTRICAL EQUIPMENT LABELING REQUIREMENTS ACCORDING TO NEC. - MEASURE THE LINE-TO-LINE AND LINE-TO-NEUTRAL VOLTAGE OF ALL SERVICE ENTRANCE CONDUCTORS PRIOR TO INSTALLING ANY SOLAR EQUIPMENT. THE VOLTAGES FOR THE 240VAC RATED.

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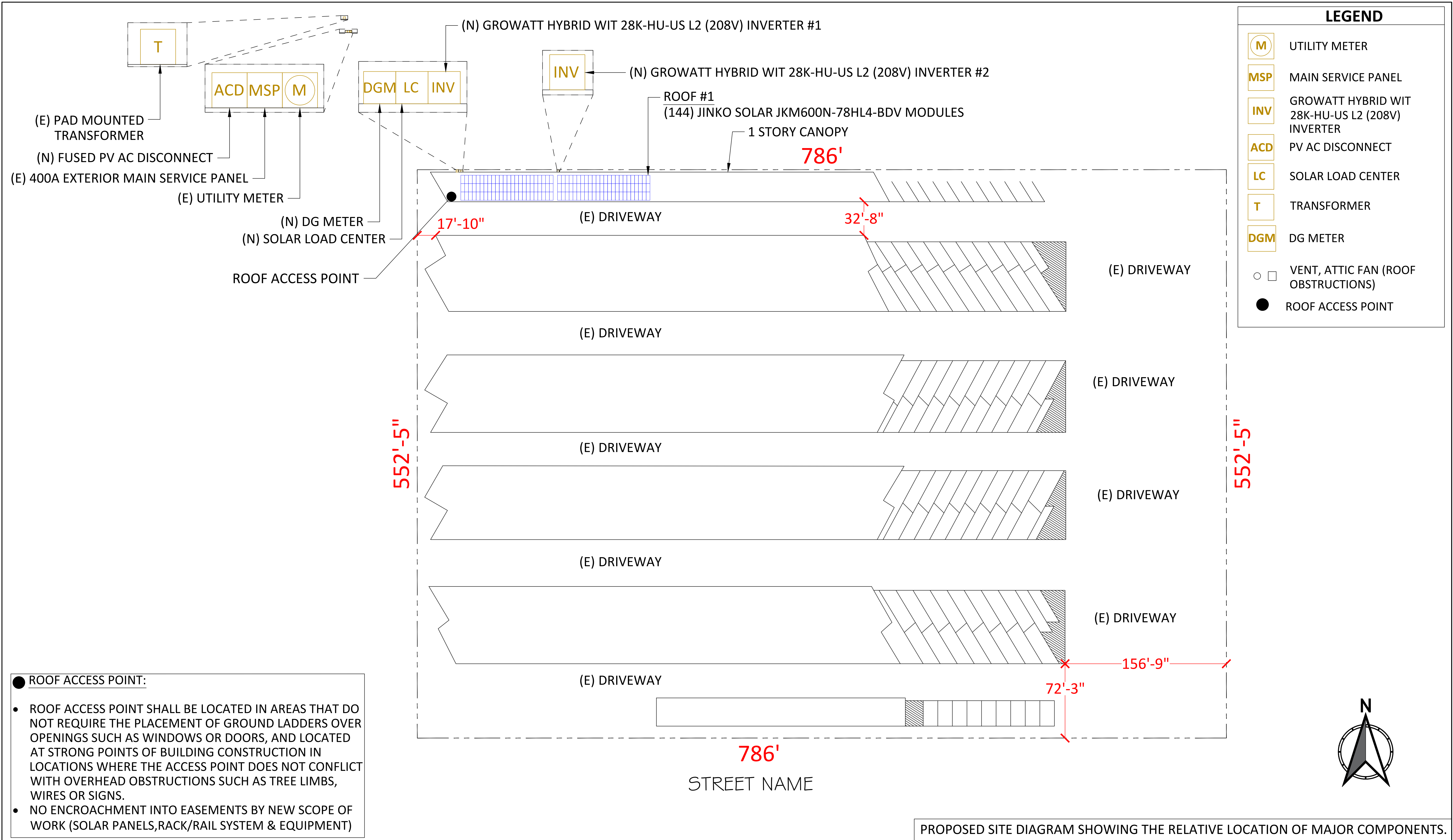
DATE	011/28/2025
PROJECT ID	SP-1
CREATED BY	RK
CHECKED BY	RS
SCALE	NTS
SHEET SIZE	24"X 36"
	ANSI D

BUILDING COMMERCIAL 12345 STREET NAME, LITCHFIELD PARK, AZ 9874 APN: 147852366325	
HOMEOWNER INFO	

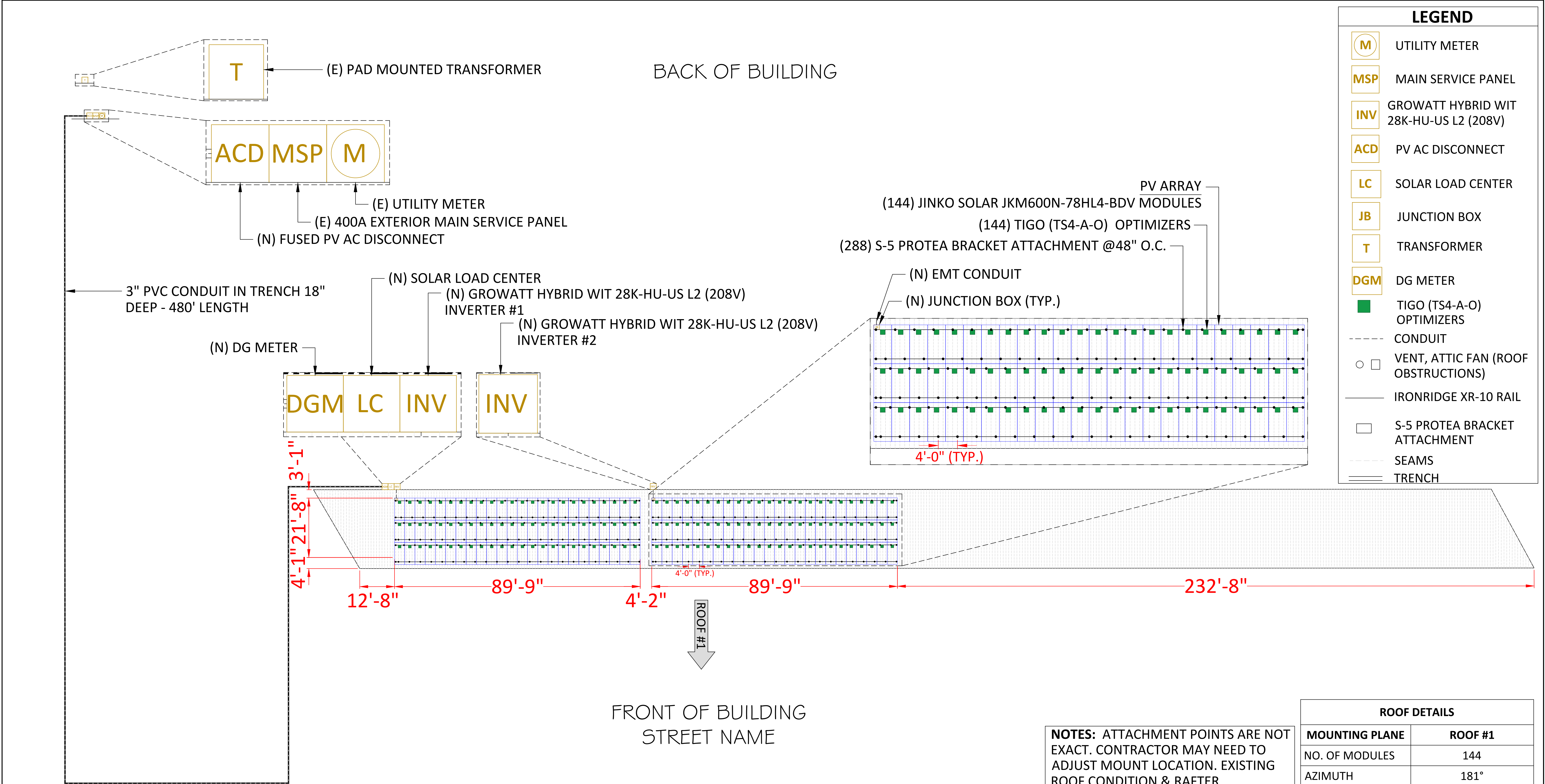
#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

PV-0
SHEET NAME
PROJECT SUMMARY
SHEET INFORMATION



	DATE	011/28/2025	BUILDING COMMERCIAL 12345 STREET NAME, LITCHFIELD PARK, AZ 9874 APN: 147852366325	#	REVISION/UPDATE	DATE		PV-1 SHEET NAME SITE PLAN SHEET INFORMATION																				
	PROJECT ID	SP-1		CREATED BY	RK	CHECKED BY			RS	SCALE	1/40"= 1'-0"	SHEET SIZE	24"X 36"	HOMEOWNER INFO				SIGNED & STAMP BY				ANSI D						
	CREATED BY	RK		CHECKED BY	RS	SCALE			1/40"= 1'-0"	SHEET SIZE	24"X 36"	HOMEOWNER INFO				SIGNED & STAMP BY				ANSI D								
	CHECKED BY	RS		SCALE	1/40"= 1'-0"	SHEET SIZE			24"X 36"	HOMEOWNER INFO				SIGNED & STAMP BY				ANSI D										
	SCALE	1/40"= 1'-0"																										
SHEET SIZE	24"X 36"	HOMEOWNER INFO				SIGNED & STAMP BY																						
		ANSI D																										



LEGEND

M

UTILITY METER

MSP

MAIN SERVICE PANEL

INV

GROWATT HYBRID WIT
28K-HU-US L2 (208V)

ACD

PV AC DISCONNECT

LC

SOLAR LOAD CENTER

JB

JUNCTION BOX

T

TRANSFORMER

DGM

DG METER

TIGO (TS4-A-O)
OPTIMIZERS

CONDUIT

○ □

VENT, ATTIC FAN (ROOF
OBSTRUCTIONS)

—

IRONRIDGE XR-10 RAIL

□

S-5 PROTEA BRACKET
ATTACHMENT

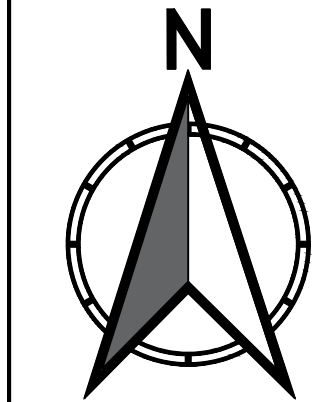
SEAMS

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TRENCH

NOTES: ATTACHMENT POINTS ARE NOT EXACT. CONTRACTOR MAY NEED TO ADJUST MOUNT LOCATION. EXISTING ROOF CONDITION & RAFTER LOCATIONS ARE SUBJECT TO FIELD VERIFICATION. ACTUAL LOCATION MAY DIFFER. INSTALL PER MANUFACTURER SPECIFICATIONS. SEE SUPPLEMENTAL DOCUMENTS FOR ATTACHMENT & STRUCTURAL DOCUMENTATION.

ROOF DETAILS	
MOUNTING PLANE	ROOF #1
NO. OF MODULES	144
AZIMUTH	181°
ROOF TILT	2°
ROOF TYPE	CORRUGATED METAL
SEAMS SPACING	12" O.C.
ARRAY AREA	4333.28
ROOF ARAY	12389.37



DATE	011/28/2025
PROJECT ID	SP-1
CREATED BY	RK
CHECKED BY	RS
SCALE	1/16" = 1'-0"
SHEET SIZE	24"X 36"
	ANSI D

BUILDING COMMERCIAL

12345 STREET NAME, LITCHFIELD PARK, AZ 9874

APN: 147852366325

HOMEOWNER INFO

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

PV-2

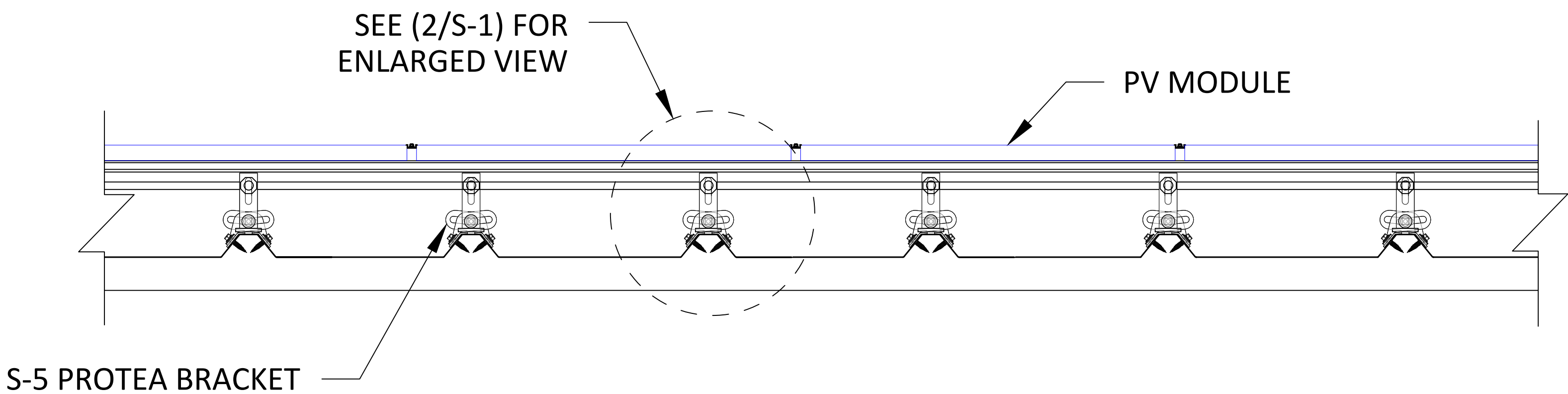
SHEET NAME

CARPORT
ATTACHMENT
DETAILS

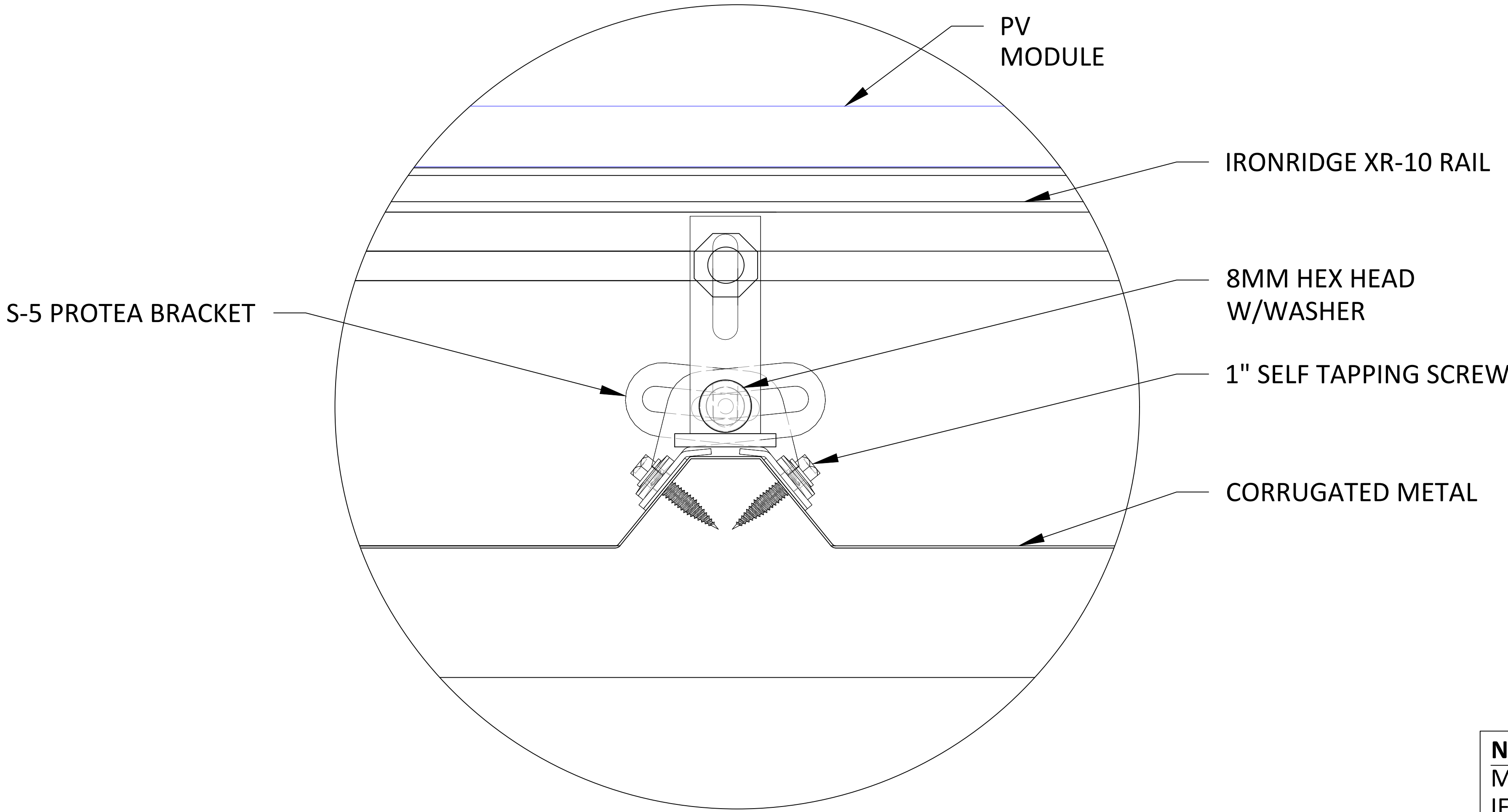
SHEET INFORMATION

STRUCTURAL NOTES:

- 1. THE SOLAR PANELS ARE TO BE MOUNTED TO THE ROOF FRAMING USING THE IRONRIDGE XR-10 SYSTEM WITH S-5 PROTEA BRACKET ATTACHMENTS.
- 2. THE MOUNTING FEET ARE TO BE SPACED AS SHOWN IN THE DETAILS, AND MUST BE STAGGERED TO ADJACENT FRAMING MEMBERS TO SPREAD OUT THE ADDITIONAL LOAD.
- 3. THE PROPOSED PV SYSTEM ADDS 2.57 PSF TO THE ROOF FRAMING SYSTEM.
- 4. ROOF LIVE LOAD = 20 PSF
- 5. GROUND SNOW LOAD = 0 lb/sqft
- 6. WIND SPEED = 101 MPH
- 7. EXPOSURE CATEGORY = B
- 8. RISK CATEGORY = II



ATTACHMENT DETAIL (SIDE VIEW)



ATTACHMENT DETAIL (ENLARGED VIEW)

NOTES: FLASHING SHALL BE APPLIED IN ACCORDANCE WITH MANUFACTURERS PRINTED INSTRUCTIONS. IF THERE IS ANY CONFLICT BETWEEN WHAT IS DEPICTED & INSTRUCTION AS PER MANUFACTURER, MANUFACTURERS INSTRUCTIONS SHALL SUPERSEDE.

NOTE: - CONTRACTOR/INSTALLER TO VERIFY COMPATIBILITY OF ANY BRANDS OR PRODUCTS SUBSTITUTED OR USED AS ALTERNATES WITHIN ANY BRAND-SPECIFIC SYSTEMS. CONTRACTOR SHALL SUPPLY AND PRESENT CERTIFICATES OF COMPATIBILITY TO THE BUILDING OFFICIAL UPON INSPECTION.

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DATE	011/28/2025
PROJECT ID	SP-1
CREATED BY	RK
CHECKED BY	RS
SCALE	NTS
SHEET SIZE	24"X 36"
	ANSI D

BUILDING COMMERCIAL 12345 STREET NAME, LITCHFIELD PARK, AZ 9874 APN: 147852366325	
HOMEOWNER INFO	

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

S-1
SHEET NAME
STRUCTURAL DETAILED DIAGRAM
SHEET INFORMATION

[DC] SYSTEM SIZE	[AC] SYSTEM SIZE
(144) MODULES x 600W	(2) INVERTERS x 28000W
86.400 kW	56.000 kW

AMBIENT TEMPERATURE SPECS	
RECORD LOW TEMP	-1°C
AMBIENT TEMP (HIGH TEMP 2%)	43°C
CONDUIT HEIGHT	7/8"
ROOF TOP TEMP	90°C
CONDUCTOR TEMPERATURE RATE	65°C
MODULE TEMPERATURE COEFFICIENT OF VOC	-0.25%/°C

SOLAR MODULE SPECIFICATIONS	
MANUFACTURER / MODEL	JINKO SOLAR JKM600N-78HL4-BDV
VMP	45.25V
IMP	13.26A
VOC	55.03V
ISC	13.87A
TEMP. COEFF. VOC	-0.25%/°C
MODULE DIMENSION	97.05 (L) x 44.65 (W)
PANEL WATTAGE	600W

INVERTER SPECIFICATIONS	
MANUFACTURER / MODEL	GROWATT HYBRID WIT 28K-HU-US L2 (208V)
NOMINAL OUTPUT VOLTAGE	208VAC
CONTINUOUS OUTPUT CURRENT	85.6A

The diagram illustrates a solar system with two parallel strings of (72) JINKO SOLAR JKM600N-78HL4-BDV modules, each consisting of 6 strings of 12 modules. These are connected to (2) GROWATT HYBRID WIT 28K-HU-US L2 (208V) inverters. The inverters feed into (N) JUNCTION BOX 01 and (N) JUNCTION BOX 02, both 600V, NEMA 3R, UL LISTED. The output of the junction boxes goes to (N) LOAD CENTER, 300A RATED, 208V, 3-PH, 4-W, NEMA 3R, UL LISTED. From the load center, the system passes through (N) DG METER, MILBANK, , 208V, NEMA 3R, UL LISTED, then through (N) AC DISCONNECT: 300A FUSED, 3-PH, 208V, NEMA 3R, UL LISTED, and finally into the main service panel. The main service panel includes (E) 400A/3P MAIN BREAKER TO HOUSE, 208V, and (E) 208V/400A BUS BAR RATING, METER-MAIN ATTACH SERVICE PANEL, 3-PH, 4-W. The system is grounded to an EXISTING GROUNDING ELECTRODE. The diagram also shows the connection to the UTILITY GRID via a UTILITY COMPANY - APS BI-DIRECTIONAL UTILITY METER# CK1990. The diagram is labeled with various components and their ratings, and includes a table of equipment details.

(72) JINKO SOLAR JKM600N-78HL4-BDV MODULES
6 STRINGS OF 12 MODULES

(N) GROWATT HYBRID WIT 28K-HU-US L2 (208V) INVERTER, 208VAC 28000WAC CONTINUOUS MAXIMUM OUTPUT CURRENT 85.6A

(N) JUNCTION BOX 01 600V, NEMA 3R, UL LISTED

(72) JINKO SOLAR JKM600N-78HL4-BDV MODULES
6 STRINGS OF 12 MODULES

(72) TIGO (TS4-A-O) OPTIMIZERS

(N) GROWATT HYBRID WIT 28K-HU-US L2 (208V) INVERTER, 208VAC 28000WAC CONTINUOUS MAXIMUM OUTPUT CURRENT 85.6A

(N) JUNCTION BOX 02 600V, NEMA 3R, UL LISTED

(N) LOAD CENTER, 300A RATED, 208V, 3-PH, 4-W, NEMA 3R, UL LISTED

(N) 110A/3P

(N) 110A/3P

(N) DG METER, MILBANK, , 208V, NEMA 3R, UL LISTED

(N) AC DISCONNECT: 300A FUSED, 3-PH, 208V, NEMA 3R, UL LISTED

(3) 250A FUSES

TO UTILITY GRID

UTILITY COMPANY - APS BI-DIRECTIONAL UTILITY METER# CK1990

POINT OF INTERCONNECT, LINE SIDE TAP

(E) 400A/3P MAIN BREAKER TO HOUSE, 208V

(E) 208V/400A BUS BAR RATING, METER-MAIN ATTACH SERVICE PANEL, 3-PH, 4-W

EXISTING GROUNDING ELECTRODE

EQUIPMENT DETAILS	
(#) PV MODULES	(144) JINKO SOLAR JKM600N-78HL4-BDV
(#) INVERTER	(2) GROWATT HYBRID WIT 28K-HU-US L2 (208V)
(#) OPTIMIZERS	(144) TIGO (TS4-A-O) OPTIMIZERS
OUTPUT CURRENT	171.20A
ROOF RACKING SYSTEM	S-5 PROTEA BRACKET /IRONRIDGE XR-10
STRING	12 STRINGS OF 12 MODULES
POINT OF INTERCONNECTION	LINE SIDE TAP IN THE EXTERIOR MAIN SERVICE PANEL WITH 400A MAIN BREAKER, 400A BUSBAR, 120/208V, 4W, 3φ
UTILITY	APS
AHJ	AVONDALE CITY

DATE	011/28/2025
PROJECT ID	SP-1
CREATED BY	RK
CHECKED BY	RS
SCALE	NTS
SHEET SIZE	24"X 36"
	ANSI D

BUILDING COMMERCIAL

12345 STREET NAME, LITCHFIELD PARK, AZ 9874

APN: 147852366325

HOMEOWNER INFO

#	REVISION/UPDATE	DATE

SIGNED & STAMP BY

E-1

SHEET NAME

SINGLE LINE DIAGRAM

SHEET INFORMATION

ID	CONDUCTOR			CONDUIT	# OF PARALLEL CIRCUITS	CURRENT-CARRYING CONDUCTORS IN CONDUIT	CONDUIT FILL PERCENT	OCPD	EGC		TEMP. CORR. FACTOR		CONDUIT FILL FACTOR	CONT. CURRENT				MAX CURRENT	BASE AMP.	DERATED AMP.	LENGTH	VOLTAGE DROP
A	10 AWG	PV WIRE		-	1	3	N/A	N/A	6 AWG	BARE COPPER	0.65	(65°C)	N/A	12.0A	x	1.25	=	15.0A	N/A	N/A	105FT	1.81%
A1	10 AWG	PV WIRE		-	1	3	N/A	N/A	6 AWG	BARE COPPER	0.65	(65°C)	N/A	12.0A	x	1.25	=	15.0A	N/A	N/A	105FT	1.81%
B	8 AWG	THWN-2	COPPER	MIN 1.25" Dia EMT	5	10	26.21%	N/A	8 AWG	THWN-2, COPPER	0.65	(65°C)	0.5	12.0A	x	1.25	=	15.0A	55A	55A x 0.5 x 0.65 = 17.9A	15FT	0.16%
B1	8 AWG	THWN-2	COPPER	MIN 1.25" Dia EMT	5	10	26.21%	N/A	8 AWG	THWN-2, COPPER	0.65	(65°C)	0.5	12.0A	x	1.25	=	15.0A	55A	55A x 0.5 x 0.65 = 17.9A	15FT	0.16%
C	1/0 AWG	THWN-2	COPPER	MIN 1.50" Dia EMT	1	4	37.06%	110A	6 AWG	THWN-2, COPPER	0.82	(43°C)	0.8	85.6A	x	1.25	=	107A	170A	170A x 0.8 x 0.82 =111.5A	92FT	1.27%
C1	1/0 AWG	THWN-2	COPPER	MIN 1.50" Dia EMT	1	4	37.06%	110A	6 AWG	THWN-2, COPPER	0.82	(43°C)	0.8	85.6A	x	1.25	=	107A	170A	170A x 0.8 x 0.82 =111.5A	5FT	0.06%
D	350 KCMIL	THWN-2	COPPER	MIN 2.50" Dia EMT	1	4	35.42%	N/A	4 AWG	THWN-2, COPPER	0.82	(43°C)	0.8	171A	x	1.25	=	214A	350A	350A x 0.8 x 0.82 =229.6A	5FT	0.05%
E	350 KCMIL	THWN-2	COPPER	MIN 3" Dia PVC	1	4	28.55%	250A	4 AWG	THWN-2, COPPER	0.82	(43°C)	0.8	171A	x	1.25	=	214A	350A	350A x 0.8 x 0.82 =229.6A	480FT	5.14%
F	350 KCMIL	THWN-2	COPPER	MIN 2.50" Dia EMT	1	4	35.42%	N/A	4 AWG	THWN-2, COPPER	0.82	(43°C)	0.8	171A	x	1.25	=	214A	350A	350A x 0.8 x 0.82 =229.6A	5FT	0.05%

	DATE	011/28/2025	BUILDING COMMERCIAL 12345 STREET NAME, LITCHFIELD PARK, AZ 9874 APN: 147852366325	#	REVISION/UPDATE	DATE	SIGNED & STAMP BY	E-2 SHEET NAME SINGLE LINE DIAGRAM
	PROJECT ID	SP-1						
	CREATED BY	RK						
	CHECKED BY	RS						
	SCALE	NTS						
	SHEET SIZE	24"X 36"						
		ANSI D	HOMEOWNER INFO					SHEET INFORMATION

1

WARNING
ELECTRIC SHOCK HAZARD
IF A GROUND FAULT IS INDICATED
NORMALLY GROUNDED CONDUCTORS
MAY BE UNGROUNDED AND ENERGIZED

LABEL LOCATION:
DC DISCONNECT, INVERTER
(PER CODE: NEC 690.5(C))
[To be used when inverter is ungrounded]

2

WARNING
ELECTRIC SHOCK HAZARD
THE DC CONDUCTORS OF THIS
PHOTOVOLTAIC SYSTEM ARE UNGROUNDED
AND MAY BE ENERGIZED

LABEL LOCATION:
DC DISCONNECT, INVERTER
(PER CODE: NEC 690.35(F))
[To be used when inverter is ungrounded]

3

WARNING
ELECTRIC SHOCK HAZARD
DO NOT TOUCH TERMINALS
TERMINALS ON BOTH LINE AND
LOAD SIDES MAY BE ENERGIZED
IN THE OPEN POSITION

LABEL LOCATION:
POINT OF INTERCONNECTION,
(PER CODE: NEC 690.17(E))

4

WARNING - Electric Shock Hazard
No user serviceable parts inside
Contact authorized service provider for assistance

LABEL LOCATION:
INVERTER, JUNCTION BOXES (ROOF),
(PER CODE: NEC690.13.G.3 & NEC 690.13.G.4)

5

WARNING: DUAL POWER SOURCE
SECOND SOURCE IS PHOTOVOLTAIC SYSTEM

LABEL LOCATION:
POINT OF INTERCONNECTION
(PER CODE: NEC 705.12(D)(4))

6

**WARNING: PHOTOVOLTAIC
POWER SOURCE**

LABEL LOCATION:
CONDUIT, COMBINER BOX
(PER CODE: NEC690.31(G)(3)(4) & NEC 690.13(G)(4)

7

PHOTOVOLTAIC SYSTEM AC DISCONNECT
RATED AC OUTPUT CURRENT 171.2 AMPS
NOMINAL OPERATING AC VOLTAGE 208 VOLTS

LABEL LOCATION:
POINT OF INTERCONNECTION,
(PER CODE: NEC 690.54)

8

WARNING
**INVERTER OUTPUT CONNECTION DO NOT
RELOCATE THIS OVERCURRENT DEVICE**

LABEL LOCATION:
POINT OF INTERCONNECTION
(PER CODE: NEC 705.12(D)(7))
[Not required if panelboard is rated not less than sum of ampere ratings
of all overcurrent devices supplying it]

9

**CAUTION: SOLAR ELECTRIC
SYSTEM CONNECTED**

LABEL LOCATION:
POINT OF INTERCONNECTION
(PER CODE: NEC690.15, 690.13(B))

10

**SOLAR PV SYSTEM EQUIPPED
WITH RAPID SHUTDOWN**

TURN RAPID SHUTDOWN
SWITCH TO THE
"OFF" POSITION TO
SHUT DOWN PV SYSTEM
AND REDUCE
SHOCK HAZARD
IN THE ARRAY

LABEL PER NEC 690.56(C)- PROVIDE AT NEW SUB
PANEL OR SERVICE PANEL FOR RAPID SHUTDOWN
COMPLIANT SYSTEM

11

CAUTION: SOLAR CIRCUIT

LABEL LOCATION:
MARKINGS PLACED ON ALL INTERIOR AND EXTERIOR DC CONDUIT, RACEWAYS,
ENCLOSURES, AND CABLE ASSEMBLIES AT LEAST EVERY 10 FT, AT TURNS AND
ABOVE/BELOW PENETRATIONS AND ALL COMBINER/JUNCTION BOXES. (PER CODE:
IFC 605.11.1.4)

12

WARNING
DISTRIBUTED GENERATION METER
PANEL MAY BE ENERGIZED BY UTILITY
OR CUSTOMER GENERATION

DG SOURCE
METER
SOCKET
UTILITY SOURCE

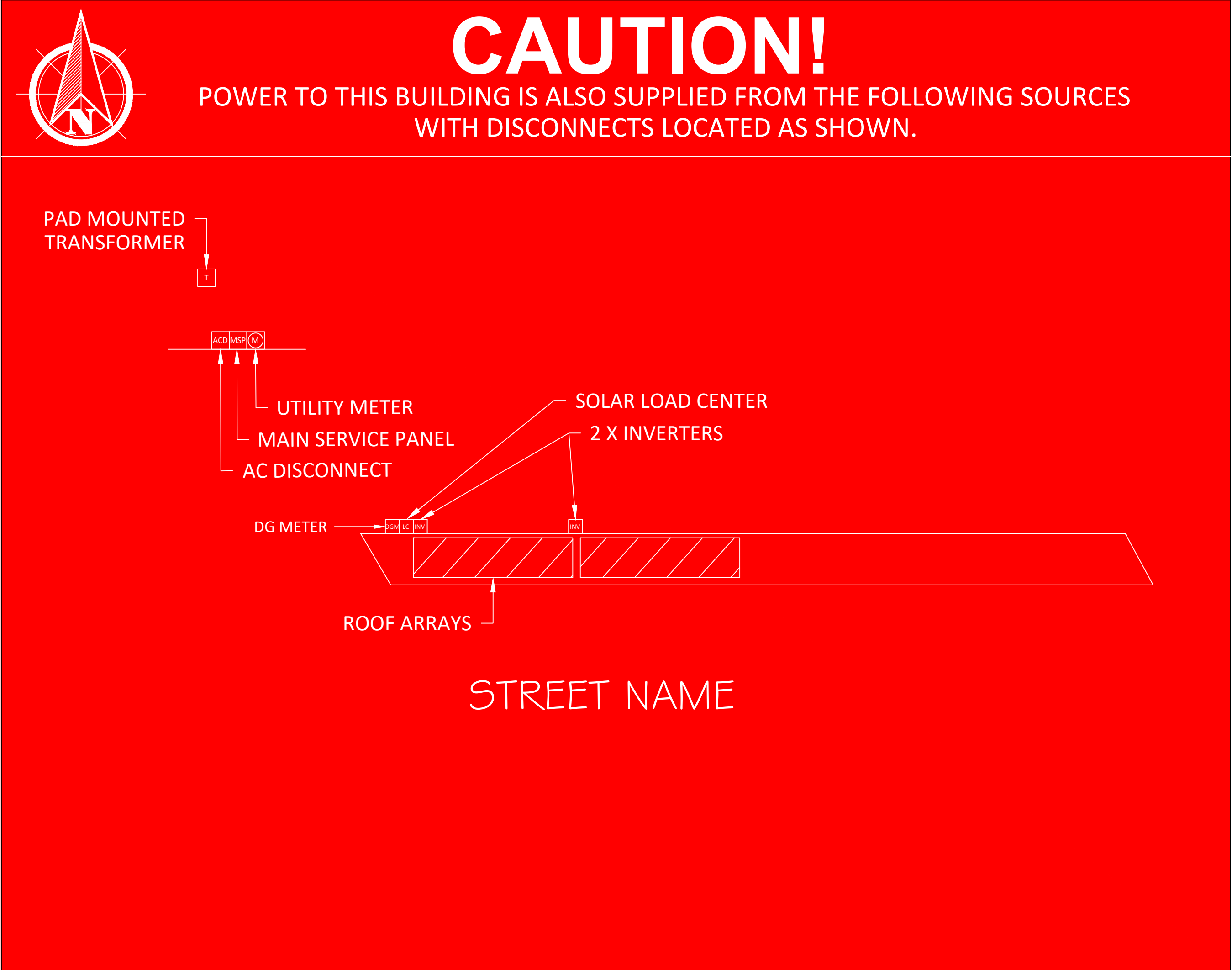
13

WARNING
ELECTRICAL SHOCK HAZARD

DISTRIBUTED GENERATION
PRESENT
TERMINALS ON LINE AND
LOAD SIDE MAY BE
ENERGIZED

14

WARNING
UTILITY DG DISCONNECT
PANEL MAY BE ENERGIZED BY
UTILITY OR CUSTOMER GENERATION



LABEL LOCATION:
EACH SERVICE EQUIPMENT LOCATION AND AT THE LOCATION(S) OF THE SYSTEM DISCONNECT(S) FOR ALL ELECTRIC POWER
PRODUCTION SOURCES CAPABLE OF BEING INTERCONNECTED

ADHESIVE FASTENED SIGNS:
• LABELS SHALL BE MADE OF RED PLASTIC MATERIAL WITH ENGRAVED WHITE LETTERS.
• LETTERS SHALL BE A MINIMUM 3/8" IN SIZE.
• THE LABELS SHALL BE PERMANENTLY ATTACHED TO THE APPROPRIATE PANEL.
• AC & DC CONDUIT, RACEWAY, ENCLOSURES, CABLE ASSEMBLIES AND JUNCTION BOXES
SHALL BE RED BACKGROUND MATERIAL WITH WHITE LETTERING MADE OF DURABLE
ADHESIVE, REFLECTIVE, WEATHER RESISTANT MATERIAL SUITABLE FOR THE
ENVIRONMENT PER UL 969; TO ALERT FIRE SERVICE TO AVOID CUTTING THEM OFF.

	DATE	011/28/2025	BUILDING COMMERCIAL 12345 STREET NAME, LITCHFIELD PARK, AZ 9874 APN: 147852366325 HOMEOWNER INFO	#	REVISION/UPDATE	DATE	SIGNED & STAMP BY	E-3 SHEET NAME PLACARDS SHEET INFORMATION
	PROJECT ID	SP-1						
	CREATED BY	RK						
	CHECKED BY	RS						
	SCALE	NTS						
	SHEET SIZE	24"X 36"						
		ANSI D						

www.jinkosolar.com



Tiger Neo N-type

78HL4-BDV

590-610 Watt

BIFACIAL MODULE WITH DUAL GLASS

N-Type

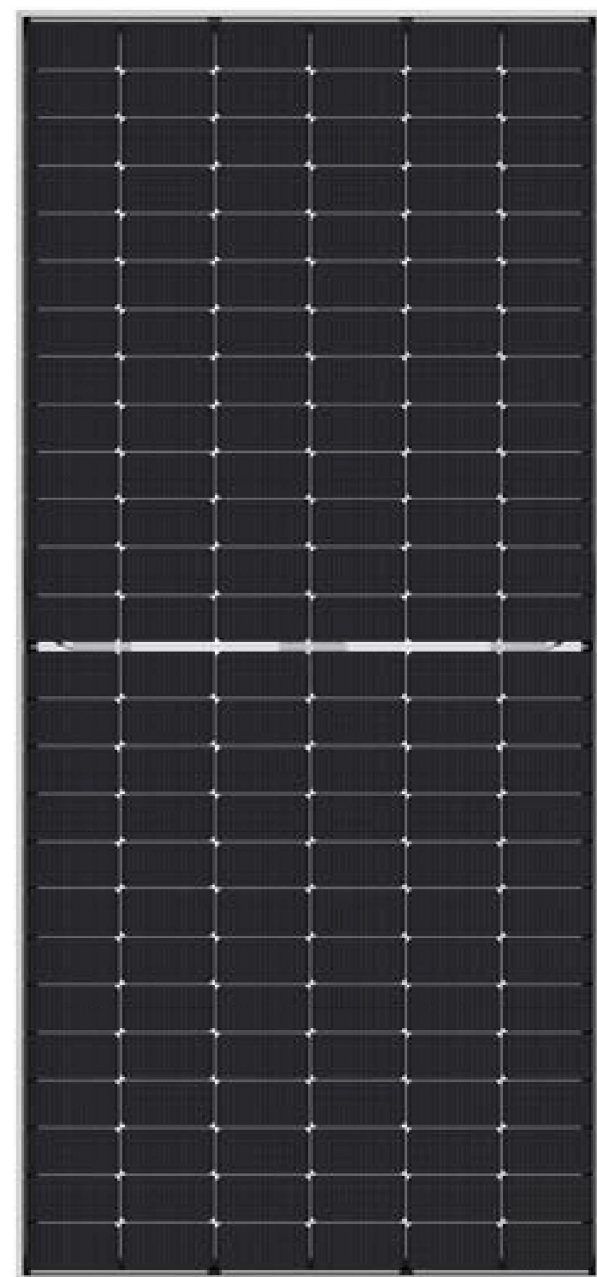
Positive power tolerance of 0~+3%

IEC61215(2016), IEC61730(2016)

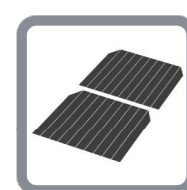
ISO9001:2015: Quality Management System

ISO14001:2015: Environment Management System

ISO45001:2018
Occupational health and safety management systems



Key Features



SMBB Technology

Better light trapping and current collection to improve module power output and reliability.



PID Resistance

Excellent Anti-PID performance guarantee via optimized mass-production process and materials control.



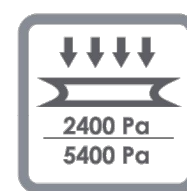
Higher Power Output

Module power increases 5-25% generally, bringing significantly lower LCOE and higher IRR.



Hot 2.0 Technology

The N-type module with Hot 2.0 technology has better reliability and lower LID/LETID.



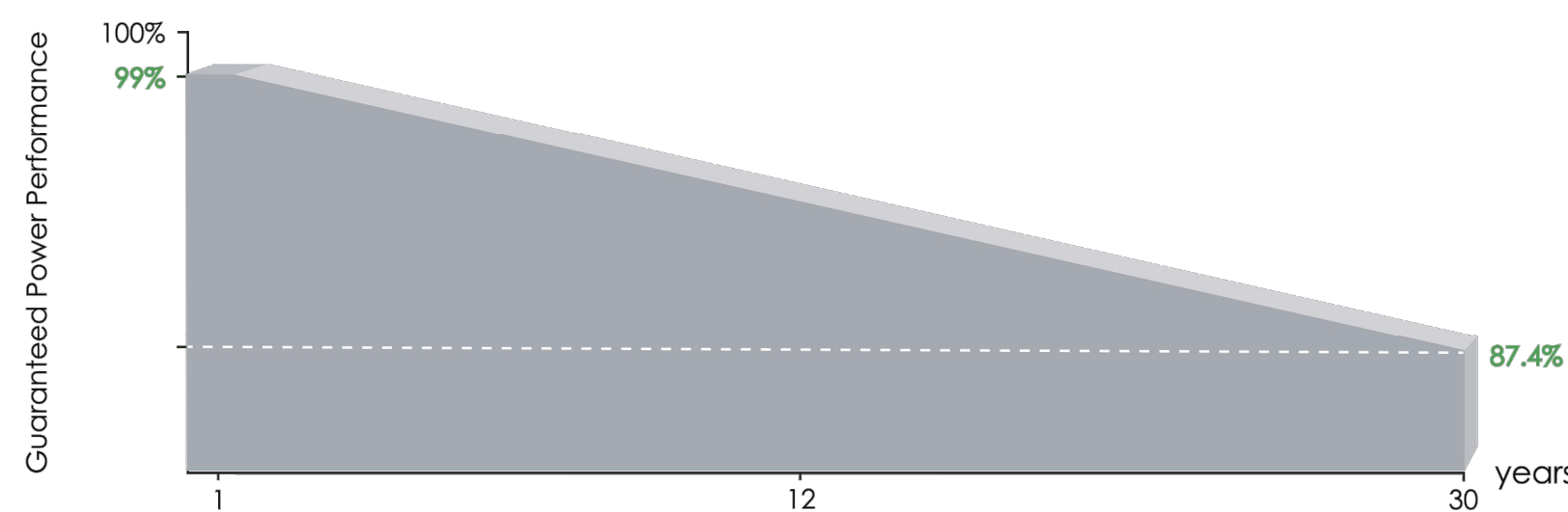
Enhanced Mechanical Load

Certified to withstand: wind load (2400 Pascal) and snow load (5400 Pascal).



Continuous Quality Assurance

LINEAR PERFORMANCE WARRANTY

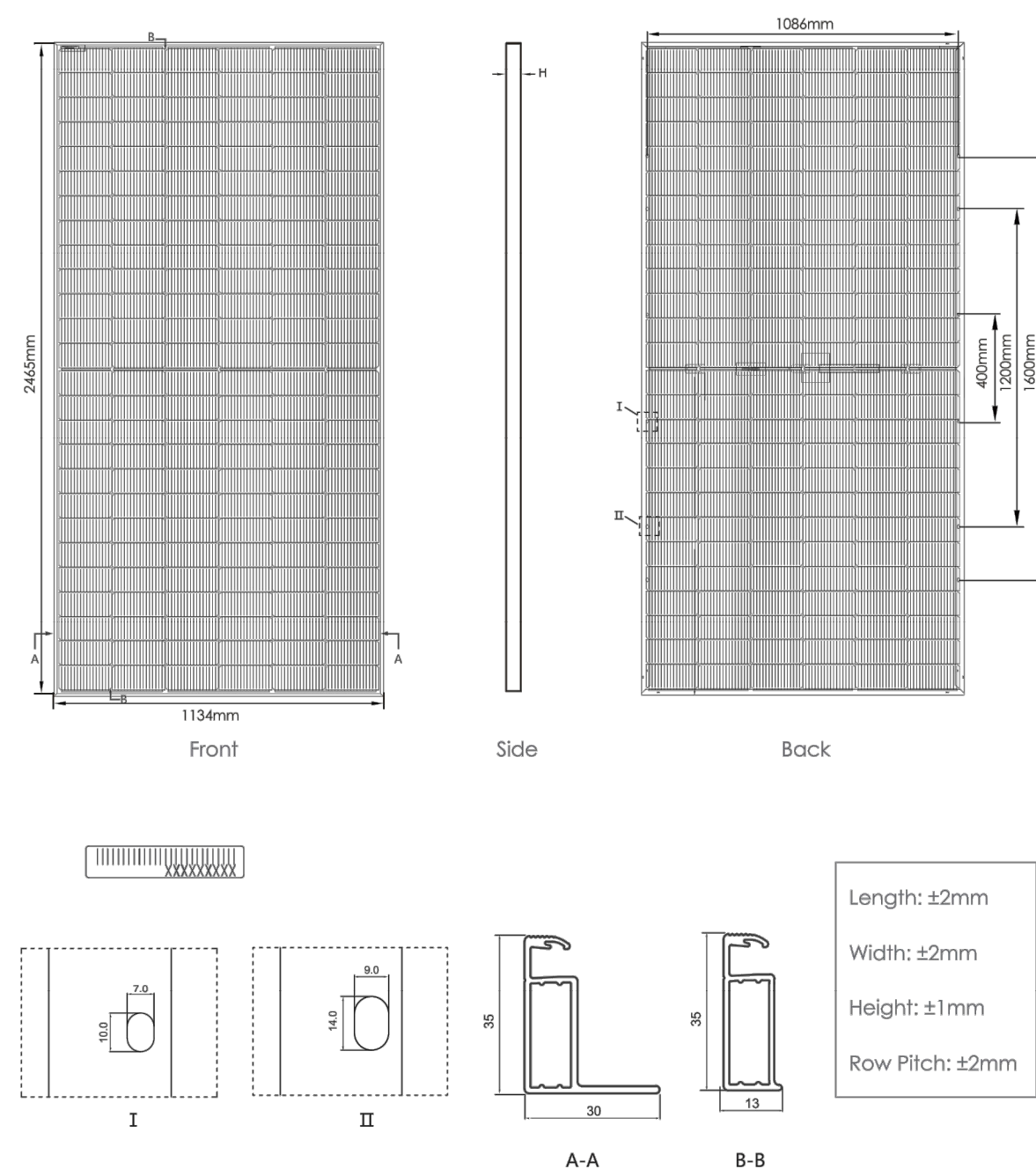


12 Year Product Warranty

30 Year Linear Power Warranty

0.40% Annual Degradation Over 30 years

Engineering Drawings



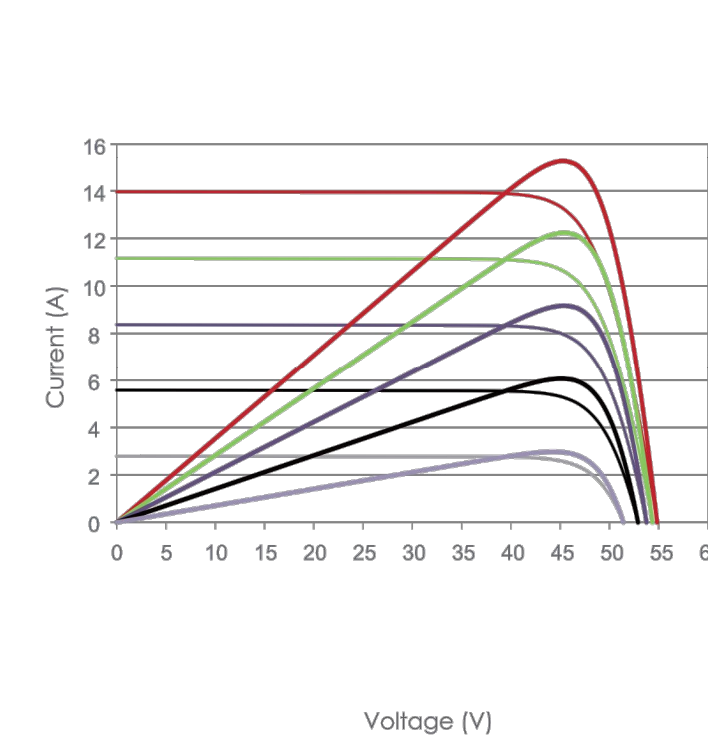
Packaging Configuration

(Two pallets = One stack)

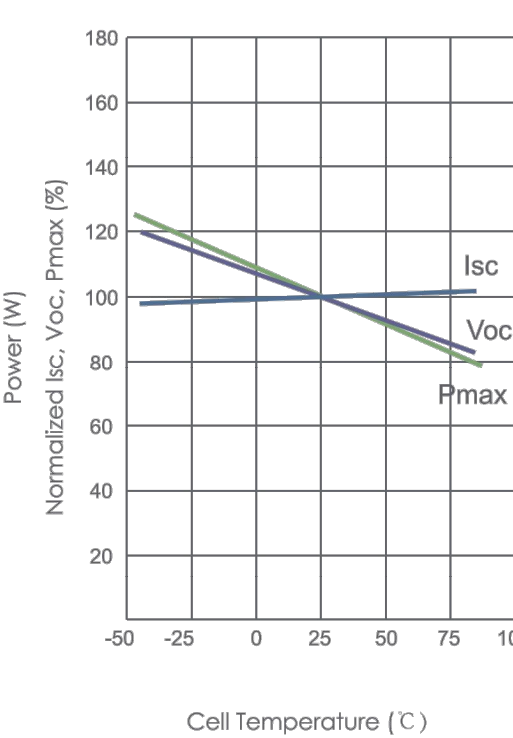
31pcs/pallets, 62pcs/stack, 496pcs/ 40'HQ Container

Electrical Performance & Temperature Dependence

Current-Voltage & Power-Voltage Curves (600W)



Temperature Dependence of Isc, Voc, Pmax



Mechanical Characteristics

Cell Type	N type Mono-crystalline
No. of cells	156 (2×78)
Dimensions	2465×1134×35mm (97.05×44.65×1.38 inch)
Weight	34.6kg (76.38 lbs)
Front Glass	2.0mm, Anti-Reflection Coating
Back Glass	2.0mm, Heat Strengthened Glass
Frame	Anodized Aluminium Alloy
Junction Box	IP68 Rated
Output Cables	TUV 1×4.0mm ² (+): 400mm , (-): 200mm or Customized Length

SPECIFICATIONS

Module Type	JKM590N-78HL4-BDV		JKM595N-78HL4-BDV		JKM600N-78HL4-BDV		JKM605N-78HL4-BDV		JKM610N-78HL4-BDV	
	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT
Maximum Power (Pmax)	590Wp	444Wp	595Wp	447Wp	600Wp	451Wp	605Wp	455Wp	610Wp	459Wp
Maximum Power Voltage (Vmp)	44.91V	41.89V	45.08V	42.00V	45.25V	42.12V	45.42V	42.23V	45.60V	42.35V
Maximum Power Current (Imp)	13.14A	10.59A	13.20A	10.65A	13.26A	10.71A	13.32A	10.77A	13.38A	10.83A
Open-circuit Voltage (Voc)	54.76V	52.02V	54.90V	52.15V	55.03V	52.27V	55.17V	52.41V	55.31V	52.54V
Short-circuit Current (Isc)	13.71A	11.07A	13.79A	11.13A	13.87A	11.20A	13.95A	11.26A	14.03A	11.33A
Module Efficiency STC (%)	21.11%		21.29%		21.46%		21.64%		21.82%	
Operating Temperature(°C)	-40°C~+85°C									
Maximum system voltage	1500VDC (IEC)									
Maximum series fuse rating	30A									
Power tolerance	0~+3%									
Temperature coefficients of Pmax	-0.30%/°C									
Temperature coefficients of Voc	-0.25%/°C									
Temperature coefficients of Isc	0.046%/°C									
Nominal operating cell temperature (NOCT)	45±2°C									
Refer. Bifacial Factor	80±5%									

BIFACIAL OUTPUT-REAR SIDE POWER GAIN

		620Wp	625Wp	630Wp	635Wp	641Wp
5%	Maximum Power (Pmax)	620Wp	625Wp	630Wp	635Wp	641Wp
	Module Efficiency STC (%)	22.16%	22.35%	22.54%	22.73%	22.91%
15%	Maximum Power (Pmax)	679Wp	684Wp	690Wp	696Wp	702Wp
	Module Efficiency STC (%)	24.27%	24.48%	24.68%	24.89%	25.10%
25%	Maximum Power (Pmax)	738Wp	744Wp	750Wp	756Wp	763Wp
	Module Efficiency STC (%)	26.38%	26.61%	26.83%	27.05%	27.28%

*STC: ☀ Irradiance 1000W/m²

📖 Cell Temperature 25°C

☁ AM=1.5

NOCT: ☀ Irradiance 800W/m²

📖 Ambient Temperature 20°C

☁ AM=1.5

🌀 Wind Speed 1m/s

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Specifications included in this datasheet are subject to change without notice.

JKM590-610N-78HL4-BDV-F1-EN (IEC 2016)

Missing or invalid reference

*File: ..\..\..\..\Bhaskar Rajak\Downloads\COMMERCIAL\Saint Elizabeth\601036-Phle-19275818-Dlabn-Ts4-A-COMMERCIAL-Saint Elizabeth
Sheet: 1*

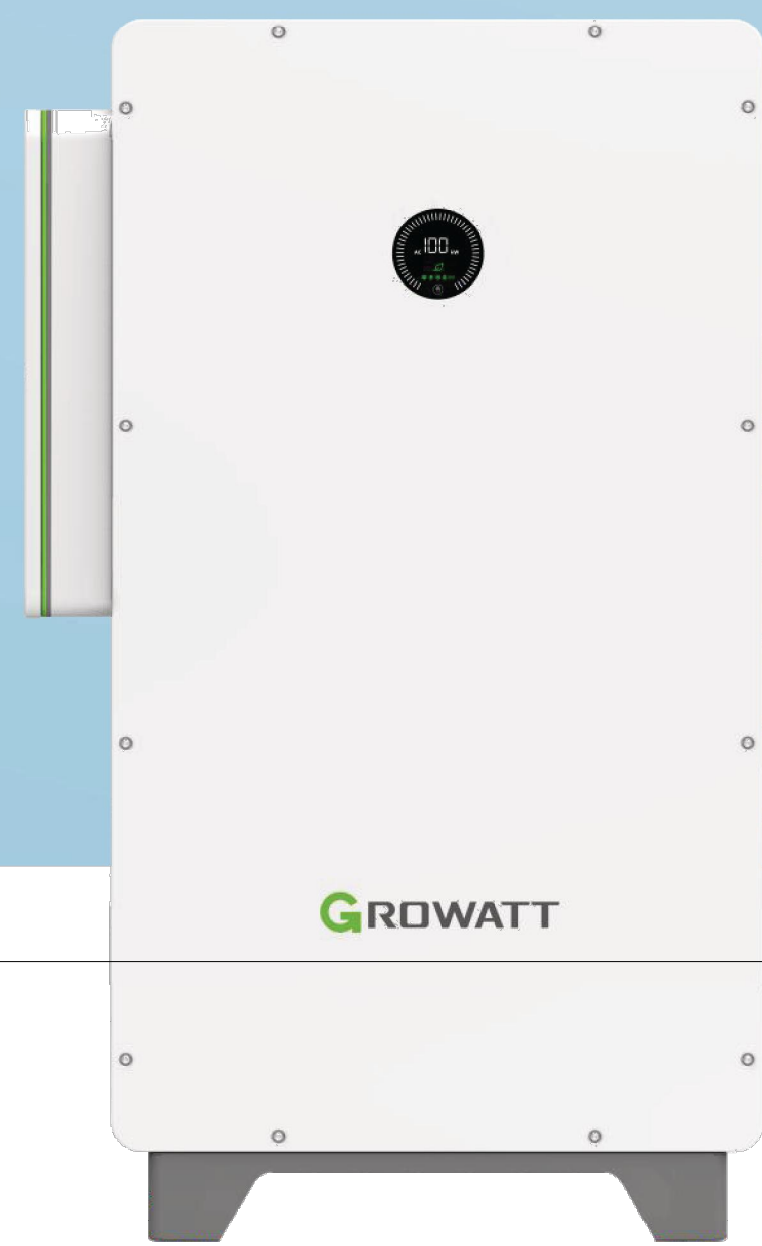
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Sheet: 2

WIT 28-55kW Hybrid Inverter

WIT 28-55K-HU-US L2

- Scalable system configuration, extend to 165kW
- Support UPS function and black start
- 100% unbalanced load when backup
- 110% continuous AC overloading capacity
- Support remote control of DG
- Grid-support functions



GROWATT

https://us.growatt.com

Preliminary Version

P O W E R
- I N G
T O M O -
R R O W

Datasheet	WIT 28K-HU-US L2		WIT 35K-HU-US L2		WIT 40K-HU-US L2		WIT 55K-HU-US L2	
Input Data (PV)								
Max. recommended PV power (for module STC)	54kW		75.6kW		86.4kW		108kW	
Start voltage	195V							
Max. input voltage	800V							
MPPT nominal voltage/range	360V, 180V-550V							
Max. input current per MPP tracker	32A							
Max. short-circuit current per MPP tracker	40A							
No. of PV strings per MPP tracker	2							
No. of MPP trackers	5		7		8		10	
Output Data (AC)								
AC nominal power/max. AC apparent power	28kW/30.8kVA		35kW/38.5kVA		40kW/44kVA		55kW/60.5kVA	
Nominal AC voltage/range	208V/220V, -15%-10%							
AC grid frequency/range	60Hz, 55-65Hz							
Max. output current	85.6A@208V 80.8A@220V		106.9A@208V 101A@220V		122.2A@208V 115.5A@220V		168.1A@208V 158.8A@220V	
Adjustable power factor	-1...+1							
THDi	<3%							
AC grid connection type	3P3W+PE/3P4W+PE							
Input Data (AC)								
AC nominal power	56kW		70kW		80kW		110kW	
Max. AC apparent power	56kVA		70kVA		80kVA		110kVA	
Nominal AC voltage/range	208V/220V, -15%-10%							
Nominal AC grid frequency/range	60Hz, 55-65Hz							
Nominal input current	155.6A@208V 147A@220V		194.4A@208V 183.7A@220V		222.2A@208V 210A@220V		305.6A@208V 288.7A@220V	
Max. input current	155.6A@208V 147A@220V		194.4A@208V 183.7A@220V		222.2A@208V 210A@220V		305.6A@208V 288.7A@220V	
Backup Power (AC power)								
AC nominal power/Max. AC apparent power	28kW/33.6kVA		35kW/42kVA		40kW/48kVA		55kW/66kVA	
Max. single phase power	9.3kW		11.6kW		13.3kW		18.3kW	
Nominal AC voltage	120V/127V(L-N), 208V/220V(L-L)							
Nominal AC frequency	60Hz							
Max. output current	93.3A@208V 88.2A@220V		116.7A@208V 110A@220V		133.3A@208V 126A@220V		183.3A@208V 173.2A@220V	
Overload capability	≤110%: Continues; 110%~120%: 10min; >120%: 200ms							
Load unbalance	100%							
THDv	<3%							
On/off grid transfer time	≤16.6ms							
Battery Data (DC)								
Battery voltage range	380V-650V, recommended @ 512V							
Max. charging / discharging current	79A		99A		113A		156A	
Continuous charging / discharging power	31.7kW		39.7kW		45.3kW		62.3kW	
BMS communication	RS485/CAN							
Types of battery	Lithium							
Efficiency								
Max. efficiency	98.2%							
Protection Devices								
PV reverse polarity protection	Yes							
Battery reverse protection	Yes							
AC/DC surge protection	Type II							
Insulation resistance monitoring	Yes							
Ground fault monitoring	Yes							
Grid monitoring	Yes							
Residual-current monitoring unit	Yes							
AC short-circuit protection	Yes							
String monitoring	Yes							
Anti-islanding protection	Yes							
PID protection	Yes							
AFCI function	Opt							
General								
Dimensions (W/H/D)	33.5/53.1/20.1inch (850/1350/510mm)							
Weight	153kg/337.3lbs				160kg/352.7lbs			
Operating temperature range	-22°F~-+140°F, >122°F derating							
Altitude	13123ft(4000m)							
Topology	Transformerless							
Cooling	Smart air cooling							
Enclosure	IP66/NEMA 4X							
Display	OLED+LED/WIFI+APP							
Interfaces:RS485/CAN	Yes							
Interfaces:WIFI//4G/LAN	Opt							
Warranty: 5 / 10 years	Yes/Opt							
UL1741:2021, UL1741 SA/SB:2020, IEEE1547:2018, IEEE1547.1:2020, HECO Rule14, E-5000, CSIP, SA17-SA18, UL1998:2018, CEC, FCC Part 15, CSA C22.2.107.1								

The right way to attach almost anything to metal roofs!

S-5!®

The Right Way

ProteaBracket™

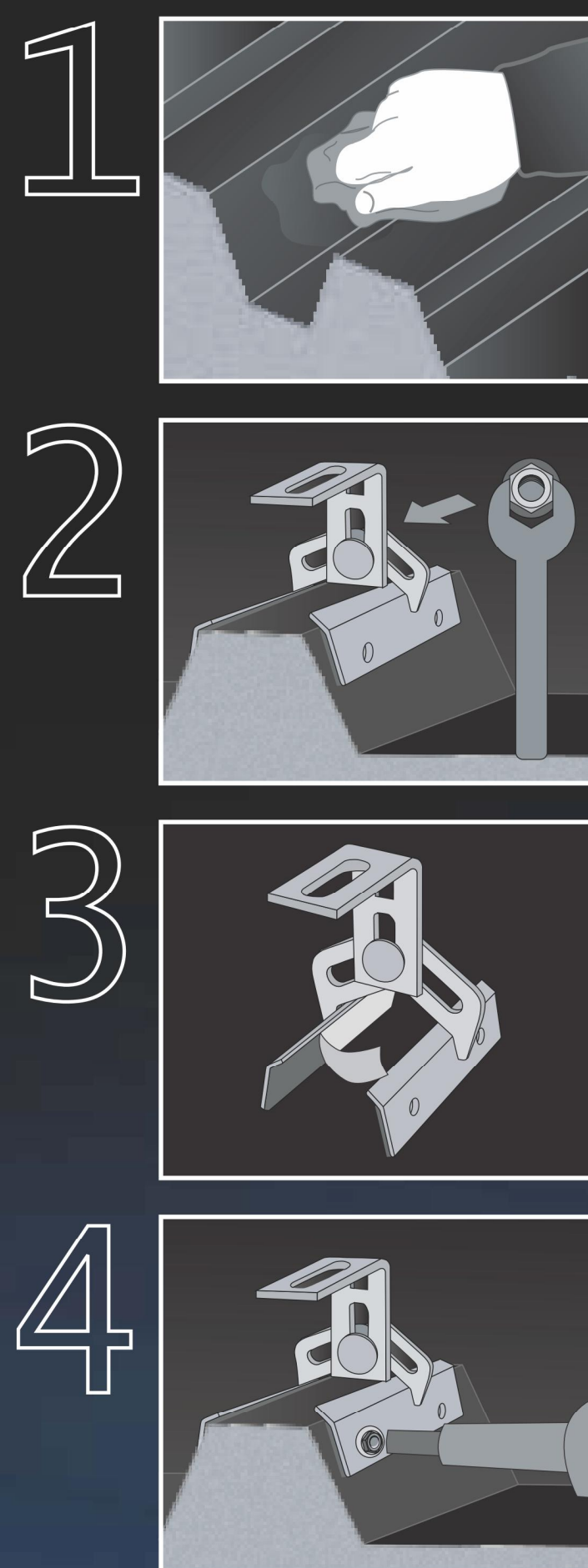
ProteaBracket™ is the most versatile attachment solution on the market, fitting most metal trapezoidal sheet profiles with and without intermediate insulation. It features an adjustable attachment base and multiple solar module attachment options (illustrated on back) to accommodate varying widths and heights. There are no messy sealants to apply and no chance for leaks; the ProteaBracket comes with factory-applied, adhesive rubber sealant to ensure quick installation and a weather-proof fit.

The ProteaBracket is mounted directly onto the crown of the panel, straddling the profile. No surface preparation is necessary; simply wipe away excess oil and debris, align, and apply. Secure ProteaBracket through all 6 pre-punched holes.

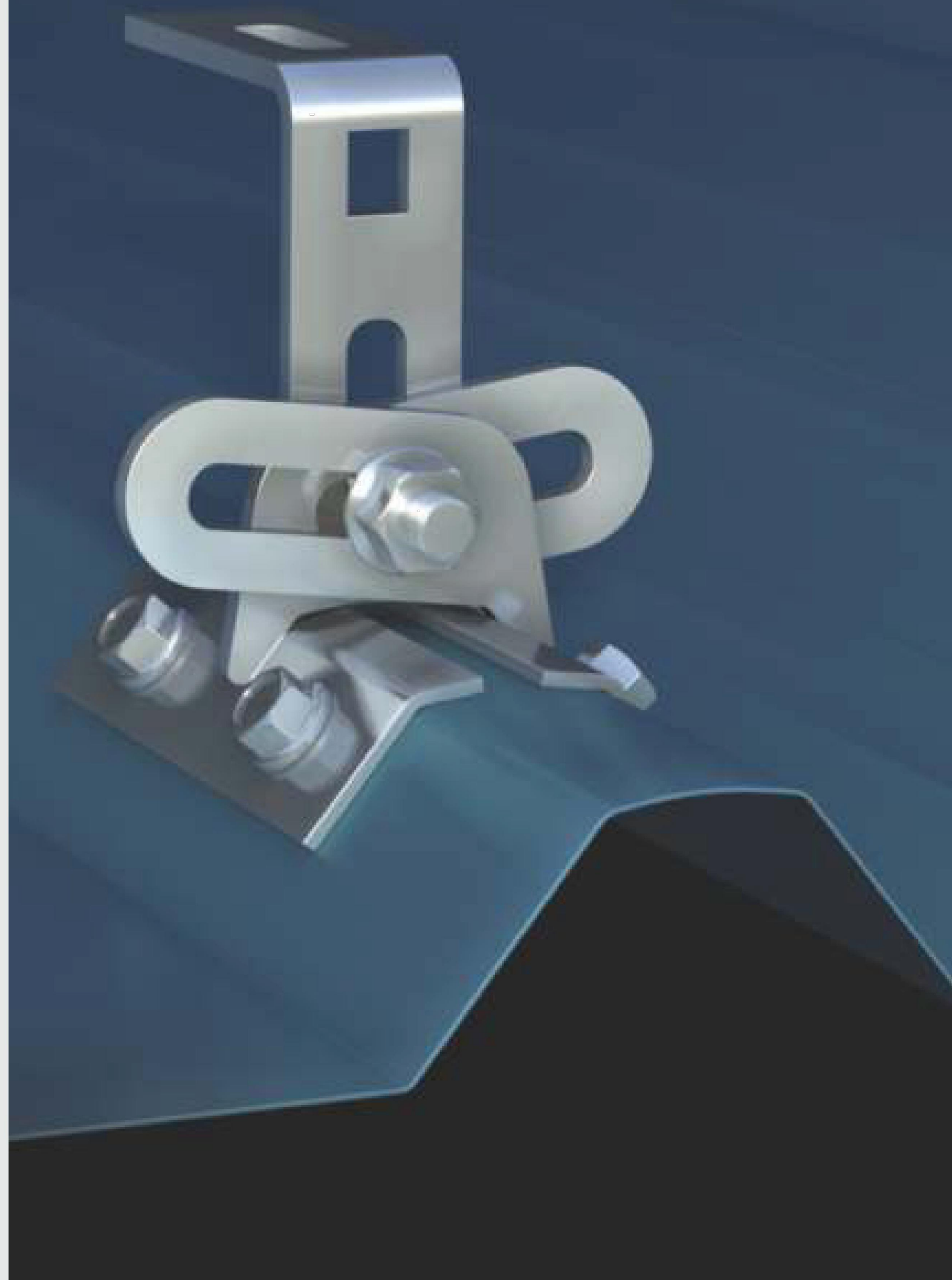
ProteaBracket is the perfect match for the S-5-PV Kit, for a solar attachment solution that is both economical and easy to use.

S-5!® ProteaBracket™ is a versatile bracket that adjusts easily to most trapezoidal roof profiles.

S-5! PV kits have an M8 bolt and are suitable for use with all S-5! clamps.



ProteaBracket™



S-5!®

The Right Way

ProteaBracket™ is the perfect solar attachment solution for most trapezoidal exposed-fastened metal roof profiles. No messy sealants to apply: the factory-applied adhesive rubber sealant weather-proofs and makes installation easy.

S-5!® holding strength is unmatched in the industry.

Each ProteaBracket™ comes with a factory-applied, adhesive rubber sealant on the base. A structural A2 stainless steel bimetal attachment bracket, ProteaBracket is compatible with most common metal roofing materials.

All four pre-punched holes must be used to achieve tested strength. For design assistance, contact Safintra South Africa (and see our website www.safintra.co.za), or visit www.S-5.com for the independent lab test data that can be used for load-critical designs and applications. Also, please visit S-5! website for more information including metallurgical compatibilities and specifications.

Multiple Attachment Options:

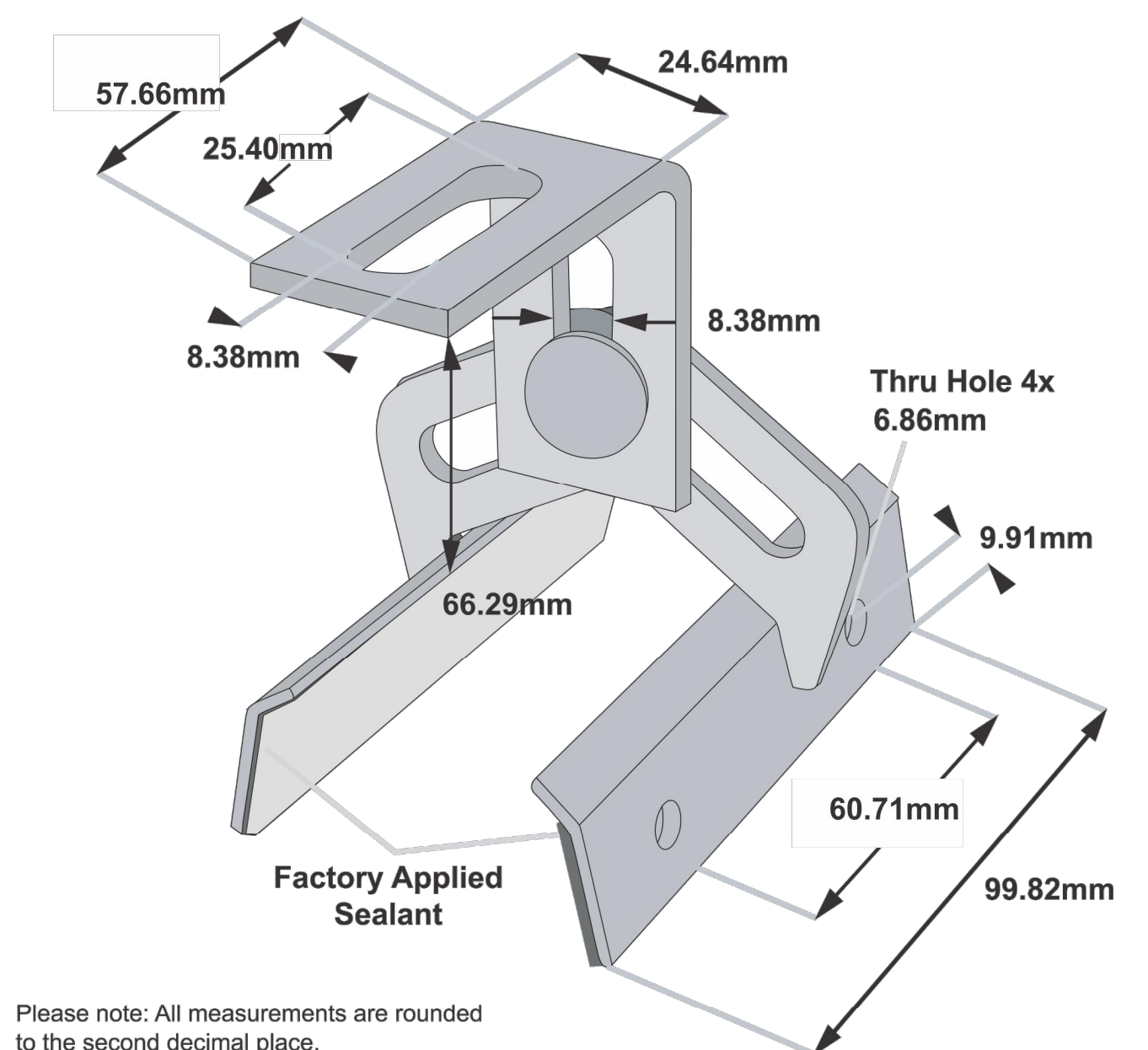
ProteaBracket™ with Top Rail option for PV attachment



ProteaBracket™ with S-5-PV Kit option (if not using a rail)



ProteaBracket™



Please note: All measurements are rounded to the second decimal place.

FLUTELINE



VERSATILE



S-5!® Warning! Please use this product responsibly!

S-5! Brackets and clamps are not tested for performance as part of a Fall Arrest or Personal Safety system. These applications need to be tested as a dynamic system and warranties or test results must be issued by the system provider. Safintra, Safal Group and its subsidiaries provide no warranties or any assurances in this application, and will accept no claims of any nature whatsoever arising out of any such applications.

Products are protected by multiple international patents. For published data regarding holding strength, bolt torque, patents and trademarks visit the S-5! website at www.S-5.com.

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Sole Agents for Africa:

SAFINTRA
world class roofing systems

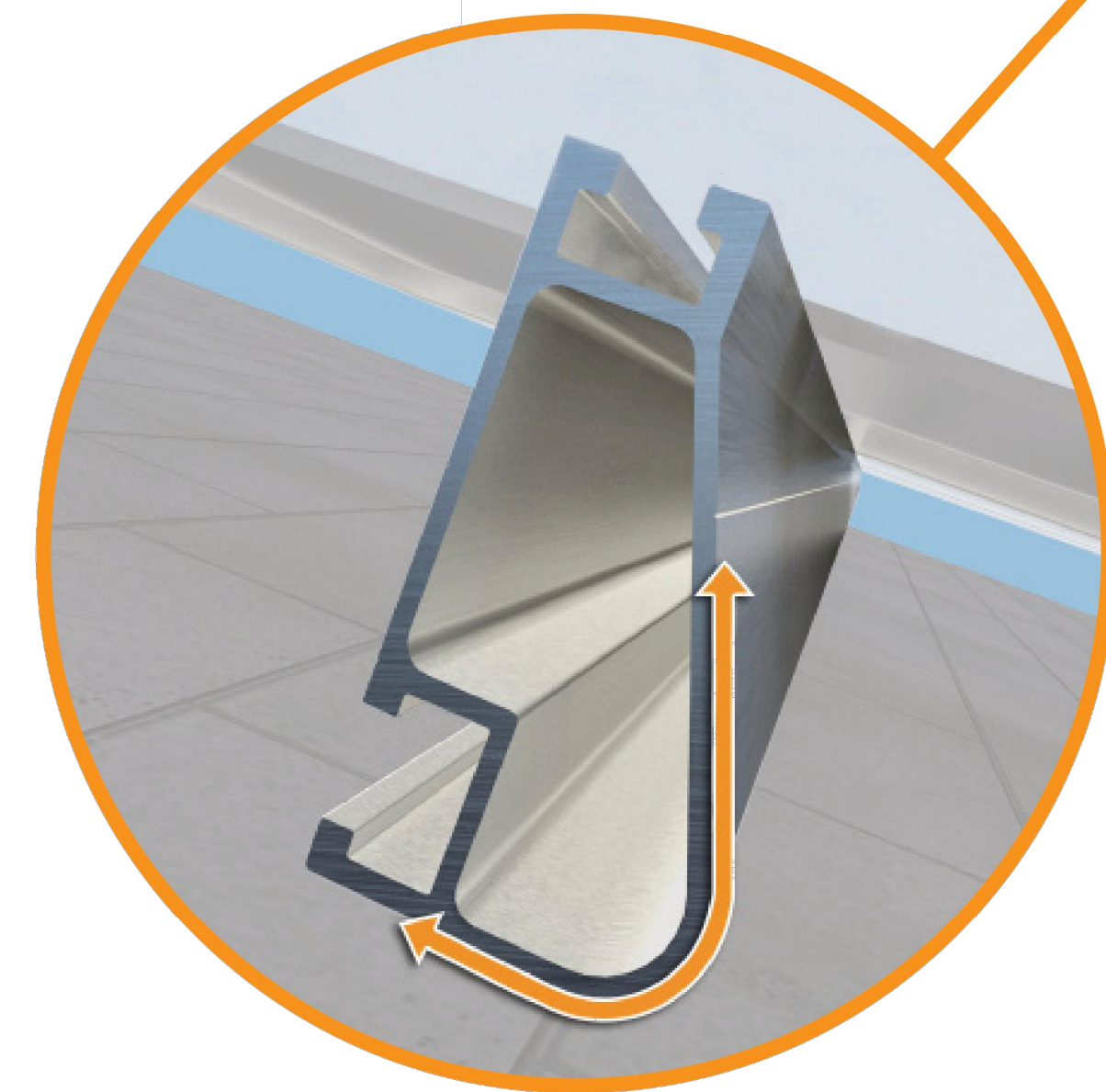
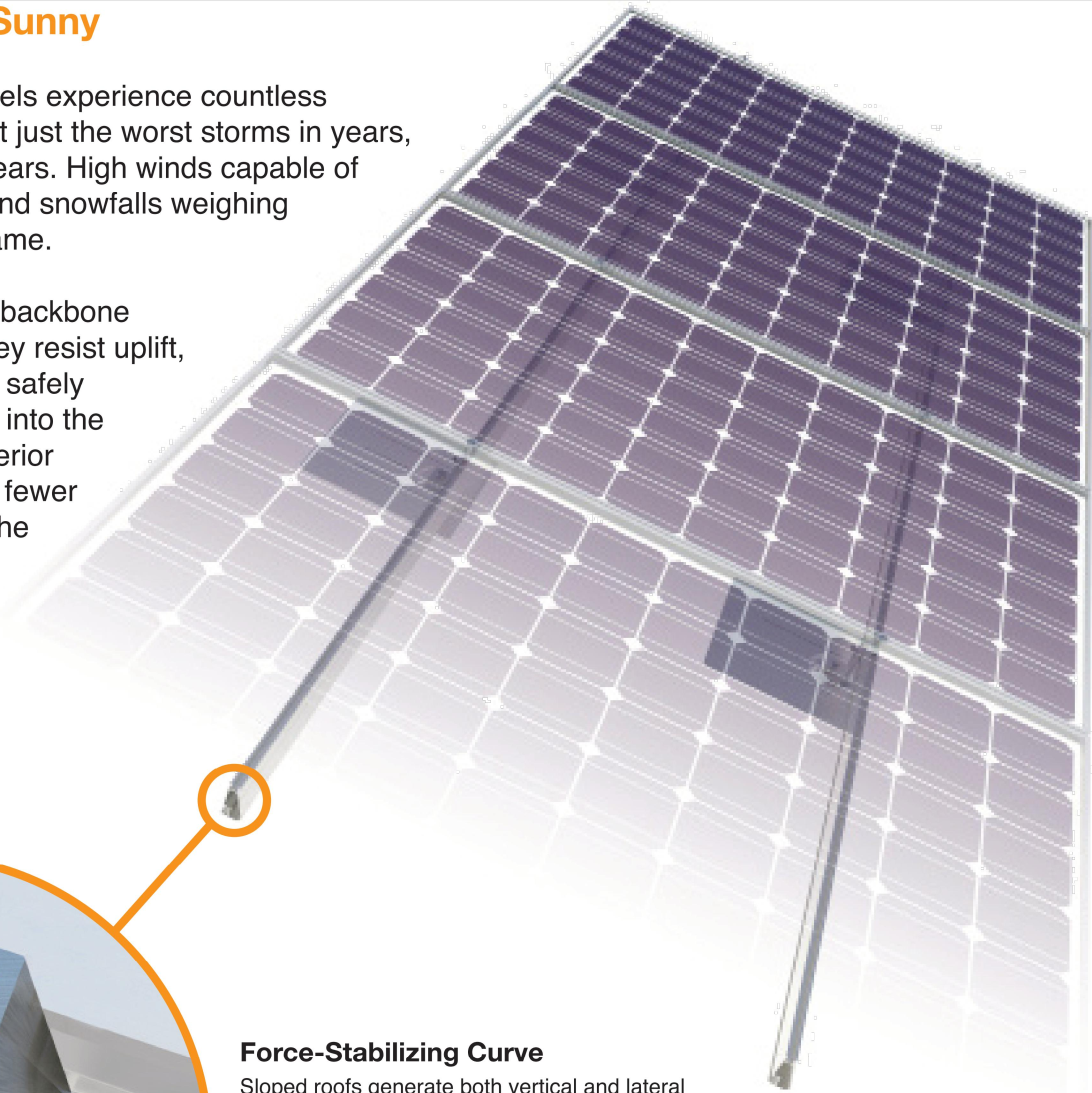
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XR Rail® Family

Solar Is Not Always Sunny

Over their lifetime, solar panels experience countless extreme weather events. Not just the worst storms in years, but the worst storms in 40 years. High winds capable of ripping panels from a roof, and snowfalls weighing enough to buckle a panel frame.

XR Rails® are the structural backbone preventing these results. They resist uplift, protect against buckling and safely and efficiently transfer loads into the building structure. Their superior spanning capability requires fewer roof attachments, reducing the number of roof penetrations and the amount of installation time.



Force-Stabilizing Curve

Sloped roofs generate both vertical and lateral forces on mounting rails which can cause them to bend and twist. The curved shape of XR Rails® is specially designed to increase strength in both directions while resisting the twisting. This unique feature ensures greater security during extreme weather and a longer system lifetime.

Compatible with Flat & Pitched Roofs



XR Rails® are compatible with FlashFoot® and other pitched roof attachments.



IronRidge® offers a range of tilt leg options for flat roof mounting applications.

Corrosion-Resistant Materials

All XR Rails® are made of 6000-series aluminum alloy, then protected with an anodized finish. Anodizing prevents surface and structural corrosion, while also providing a more attractive appearance.



XR Rail® Family

The XR Rail® Family offers the strength of a curved rail in three targeted sizes. Each size supports specific design loads, while minimizing material costs. Depending on your location, there is an XR Rail® to match.



XR10

XR10 is a sleek, low-profile mounting rail, designed for regions with light or no snow. It achieves spans up to 6 feet, while remaining light and economical.

- 6' spanning capability
- Moderate load capability
- Clear & black anodized finish
- Internal splices available



XR100

XR100 is the ultimate residential mounting rail. It supports a range of wind and snow conditions, while also maximizing spans up to 10 feet.

- 10' spanning capability
- Heavy load capability
- Clear & black anodized finish
- Internal splices available



XR1000

XR1000 is a heavyweight among solar mounting rails. It's built to handle extreme climates and spans up to 12 feet for commercial applications.

- 12' spanning capability
- Extreme load capability
- Clear anodized finish
- Internal splices available

Rail Selection

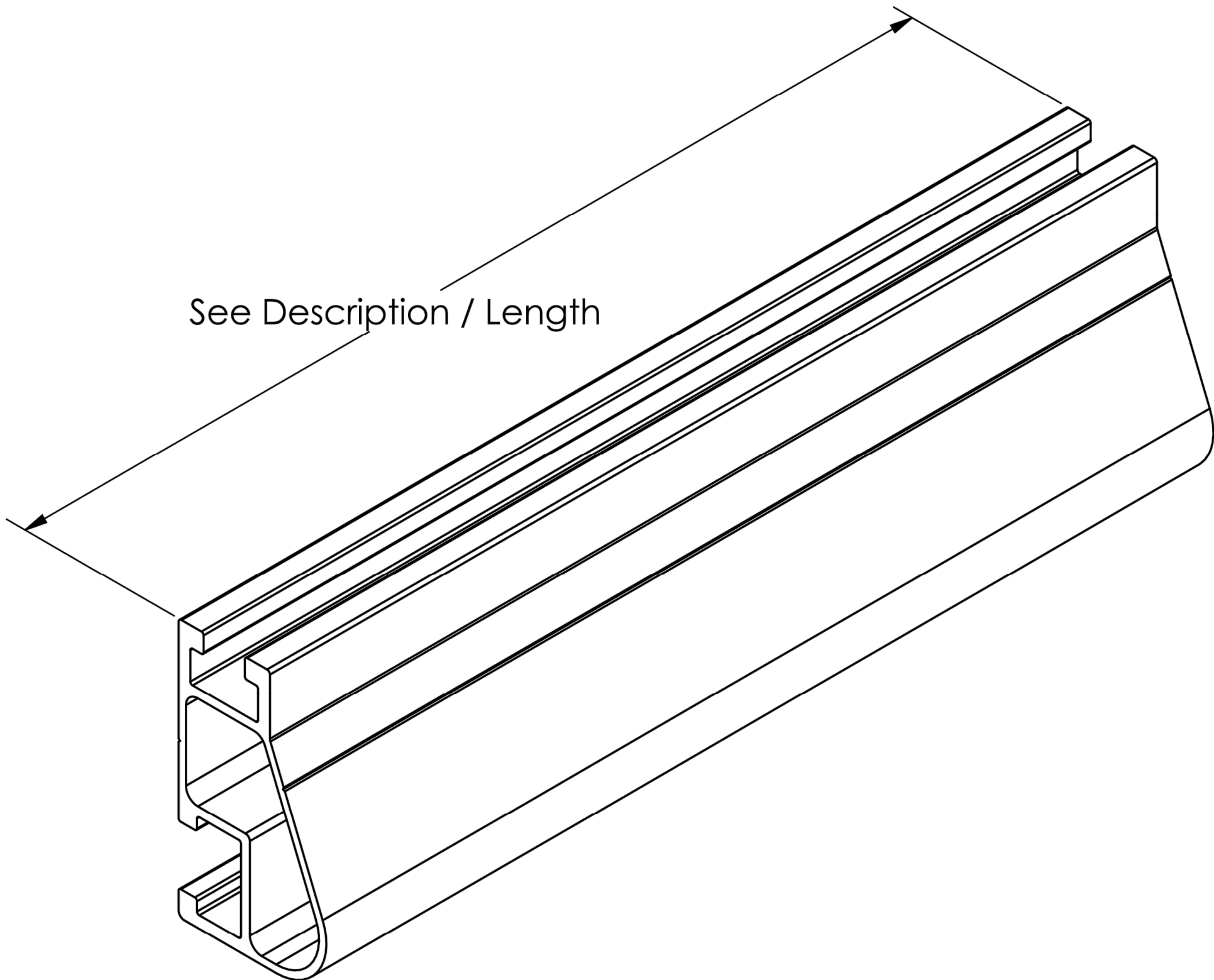
The table below was prepared in compliance with applicable engineering codes and standards.* Values are based on the following criteria: ASCE 7-16, Gable Roof Flush Mount, Roof Zones 1 & 2e, Exposure B, Roof Slope of 8 to 20 degrees and Mean Building Height of 30 ft. Visit IronRidge.com for detailed certification letters.

Load		Rail Span					
Snow (PSF)	Wind (MPH)	4'	5' 4"	6'	8'	10'	12'
None	90	XR10		XR100		XR1000	
	120						
	140						
	160						
20	90						
	120						
	140						
	160						
30	90						
	160						
40	90						
	160						
80	160						
120	160						

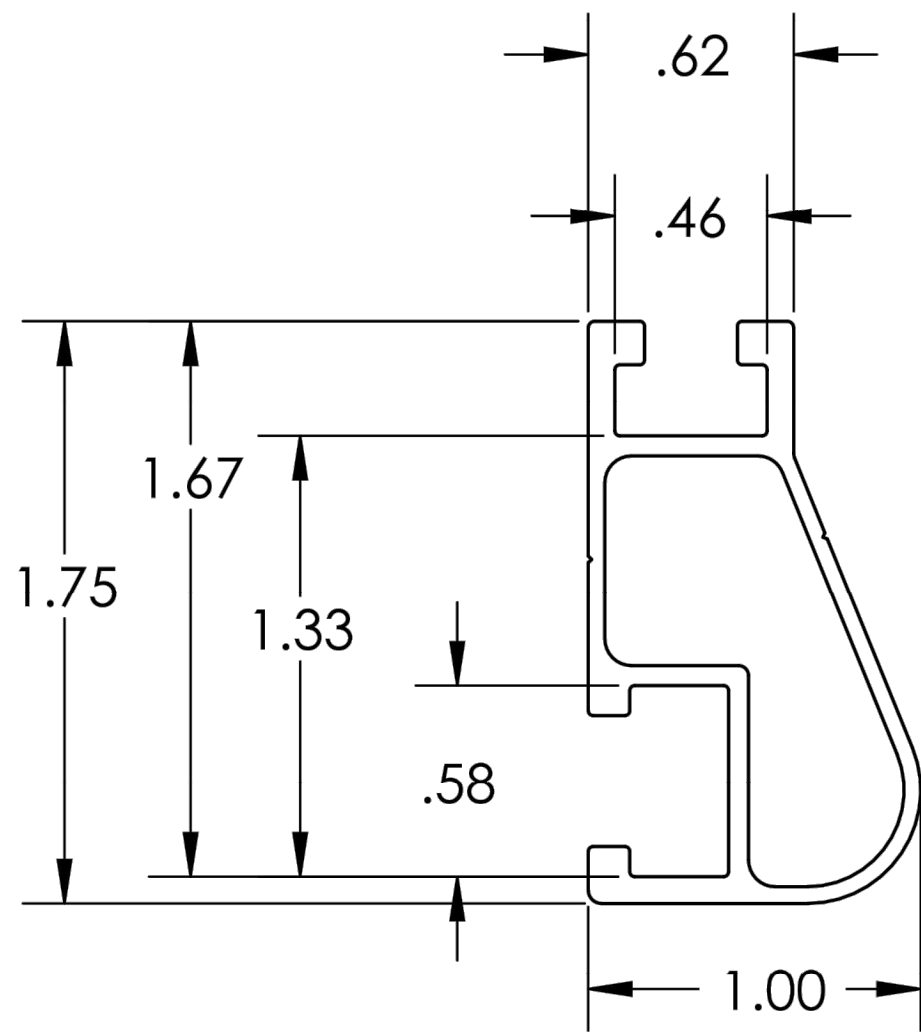
*Table is meant to be a simplified span chart for conveying general rail capabilities. Use approved certification letters for actual design guidance.



XR10® Rail



See Description / Length



Rail Section Properties	
Property	Value
Total Cross-Sectional Area	0.363 in ²
Section Modulus (X-axis)	0.136 in ³
Moment of Inertia (X-axis)	0.124 in ⁴
Moment of Inertia (Y-axis)	0.032 in ⁴
Torsional Constant	0.076 in ³
Polar Moment of Inertia	0.033 in ⁴

Clear Part Number	Black Part Number	Description / Length	Material	Weight
XR-10-132A	XR-10-132B	XR10, Rail 132" (11 Feet)	6000-Series Aluminum	4.67 lbs.
XR-10-168A	XR-10-168B	XR10, Rail 168" (14 Feet)		5.95 lbs.
XR-10-204A	XR-10-204B	XR10, Rail 204" (17 Feet)		7.22 lbs.